Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Introduction to Planetary Science

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Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning Objectives

By the end of this lecture, you will be able to:

- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above T_{eff}
- Evaluate atmospheric escape mechanisms and retention criteria

Recap: Lecture 4

- Metal–silicate separation driven by magma oceans
- Hf–W chronometry: Earth's core formed in ~30–60 Myr
- Goldschmidt classification: siderophile, lithophile, chalcophile, atmophile
- Dynamo requirements: conducting fluid + convection + $\mathrm{Rm} \gtrsim 10$–100
- Mars lost its dynamo ~4.1 Ga → atmospheric erosion

Today: The thin gaseous envelope that controls a planet's surface conditions.

1. Atmospheric Composition

Earth’s atmosphere, ISS (2009). Credit: NASA

Primary Atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- Requires mass $\gtrsim$ 5–10 $M_\oplus$ to capture nebular gas
- Disk disperses in ~3–10 Myr (Lecture 2)
- **Gas giants:** Jupiter and Saturn, massive $\mathrm{H_2}$/He envelopes
- **Ice giants:** Uranus and Neptune, envelopes only 10–20% of total mass
- Terrestrial planets too small → any primordial H quickly lost

Secondary Atmospheres

Produced by outgassing from the interior.

- Volcanism and magma ocean degassing release dissolved volatiles
- Speciation depends on **oxygen fugacity** (Lecture 4):
  - Oxidising: CO_2, H_2O, N_2
  - Reducing: H_2, CO, N_2
- **Venus:** CO_2-dominated (96.5%), 92 bar
- **Mars:** CO_2-dominated (95%), but only 6 mbar
- **Titan:** thick N_2 (from NH_3 conversion), 1.5 bar

Where Do Atmospheres Come From?

A pictorial summary of the three classes.

Primary Directly captured nebular gas; H_2/He dominated. Requires $M \gtrsim 5\text{–}10\,M_\oplus$ before disk dispersal.

Secondary Outgassed from the interior during accretion + cooling. Composition set by f_{O_2} of the mantle.

Tertiary Modified by surface processes, photochemistry, and (for Earth) biology.

Over time, a planet's atmospheric inventory can be **replaced** or **re-equilibrated** entirely: the present composition may not reflect the initial inventory.

Tertiary Atmospheres

Modified by surface processes, photochemistry, or biology.

- Earth's original outgassed atmosphere: likely $\mathrm{CO_2}$ + $\mathrm{N_2}$ (similar to Venus)
- Oxygenic photosynthesis transformed it over billions of years
- Present atmosphere: $\mathrm{N_2}$ (78%) + $\mathrm{O_2}$ (21%)
- $\mathrm{O_2}$ **is entirely biogenic**, and would disappear in ~Myr without life

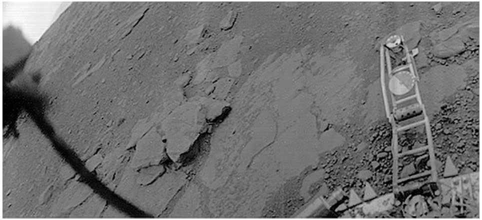
Earth's atmosphere is a **tertiary atmosphere**: the product of billions of years of biological modification.

Comparative Atmospheric Properties

	Venus	**Earth**	**Mars**	**Jupiter**	**Titan**
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	735	288	210	n/a	94
Dominant gas	$\mathrm{CO_2}$	$\mathrm{N_2}$	$\mathrm{CO_2}$	$\mathrm{H_2}$	$\mathrm{N_2}$
Secondary gas	$\mathrm{N_2}$	$\mathrm{O_2}$	$\mathrm{N_2}$	He	$\mathrm{CH_4}$
μ	43.4	29.0	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 735 K.

Venus: A Secondary Atmosphere in Extremis



Credit: USSR/NASA NSSDC, public domain

- *Venera 13* lander, 1 March 1982
- Surface pressure: 92 bar
- Surface temperature: 735 K
- Orange sky: thick $\mathrm{CO_2}$ atmosphere scatters light
- Lander survived 127 minutes

A vivid demonstration of how a massive secondary atmosphere transforms surface conditions.

2. Hydrostatic Equilibrium

Earth’s atmosphere. Credit: NASA/ISS

Hydrostatic Equilibrium

Consider a thin slab of atmosphere at height z, thickness $\mathrm{d}z$, area A:

- Pressure from below (upward): $P(z) \cdot A$
- Pressure from above (downward): $P(z + \mathrm{d}z) \cdot A$
- Weight (downward): $\rho(z)\, g\, A\, \mathrm{d}z$

Force balance:

$$P(z)\, A - P(z + \mathrm{d}z)\, A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Pressure decreases with altitude: each layer supports the weight above it.

The Ideal Gas Law (Atmospheric Form)

To solve hydrostatic equilibrium, we need $\rho(P)$:

$$P = n\, k_\mathrm{B} T = \frac{\rho\, k_\mathrm{B} T}{\mu\, m_u}$$

n	number density
k_B	Boltzmann constant (1.381×10^{-23} J K^{-1})
μ	mean molecular weight (dimensionless, in amu)
m_u	atomic mass unit (1.661×10^{-27} kg)

Rearranging: $\rho = P\, \mu\, m_u / (k_\mathrm{B} T)$

The Barometric Formula

Substitute $\rho = P\mu m_u/(k_\mathrm{B}T)$ into hydrostatic equilibrium:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B}T}\, P$$

For an **isothermal** atmosphere (constant T, constant g), this separates:

$$\boxed{P(z) = P_0\, \exp\left(-\frac{z}{H}\right)}$$

where $H = k_\mathrm{B}T/(\mu\, m_u\, g)$ is the **pressure scale height**.

- Pressure drops by $e \approx 2.718$ every scale height
- 99% of the atmospheric mass lies below ~ $5H$

3. Blackboard Derivation: Scale Height

Earth’s atmosphere. Credit: NASA/ISS

Deriving the Scale Height

Goal: Derive $H = k_\mathrm{B}T/(\mu\, m_u\, g)$ from hydrostatic equilibrium + ideal gas law, and compute H for solar system bodies.

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_\mathrm{B} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\, \mu\, m_u}{k_\mathrm{B} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B} T} P \equiv -\frac{P}{H}$$

The Atmospheric Scale Height

$$\boxed{H = \frac{k_\mathrm{B} T}{\mu\, m_u\, g}}$$

Physical interpretation:

- **Higher** T → larger H: hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H: harder to loft
- **Stronger gravity** g → smaller H: more compressed

Key insight: H encodes the competition between thermal energy and gravitational binding.

Worked Example: Earth's Scale Height

$$H_\oplus = \frac{k_\mathrm{B} T}{\mu\, m_u\, g} = \frac{1.381 \times 10^{-23} \times 288}{28.97 \times 1.661 \times 10^{-27} \times 9.81}$$

$$= \frac{3.98 \times 10^{-21}}{4.72 \times 10^{-25}} \approx 8400\ \mathrm{m} \approx 8.4\ \mathrm{km}$$

- Commercial aircraft: 10–12 km (~1.2–1.4 H), pressure ~0.2–0.3 atm
- Top of Everest: 8.8 km (~1 H), pressure ~0.3 atm
- 99% of mass below ~42 km (~5 H)

Scale Heights Across the Solar System

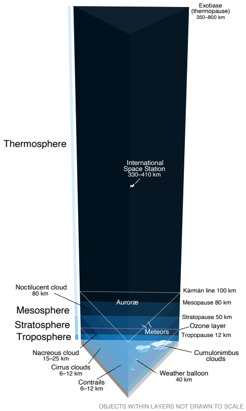
Body	T **(K)**	μ	g **(m s^{-2})**	H **(km)**
Venus	735	43.4	8.87	15.8
Earth	288	29.0	9.81	8.5
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	27
Titan	94	28.6	1.35	21

- Jupiter: large H despite strong gravity, because $\mathrm{H_2}$ is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot but heavy molecules + decent gravity → moderate H

4. Vertical Structure

Earth’s atmospheric layers. Credit: Wikimedia, public domain

Atmospheric Layers



Credit: Wikimedia, public domain

Layers defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$:

1. **Troposphere** (0–12 km): T decreases with z
   Heated from below; convective
2. **Stratosphere** (12–50 km): T increases
   UV absorbed by ozone layer
3. **Mesosphere** (50–85 km): T decreases
   Coldest point at mesopause (~190 K)
4. **Thermosphere** (>85 km): T increases
   EUV absorption; T > 1000 K but gas is rarefied

The Adiabatic Lapse Rate

In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8\,\mathrm{K\,km^{-1}}$
- Observed average: $\sim 6.5\,\mathrm{K\,km^{-1}}$ (lower due to latent heat from water condensation)
- Physical picture: rising air expands, does work on surroundings, cools

Tropopause height: ~8 km (poles) to ~17 km (equator).

Comparative Vertical Structures

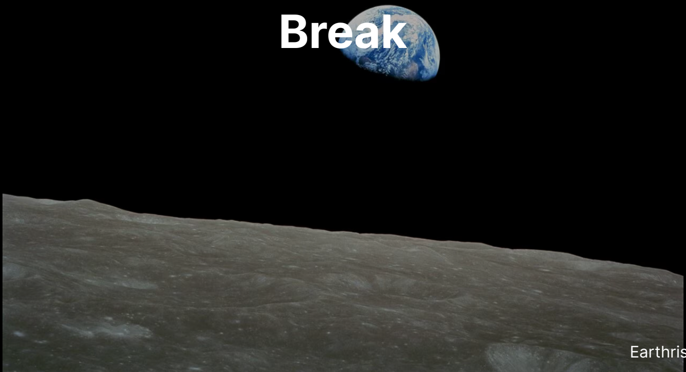
Venus: Massive troposphere (~65 km). No ozone layer → no stratospheric inversion. Cloud deck at 48–70 km.

Mars: Thin troposphere (~40 km), directly into thermosphere. No significant ozone. Dust dominates atmospheric heating.

Jupiter: Troposphere extends hundreds of km deep. Stratosphere heated by CH_4. No solid surface; pressure increases continuously.

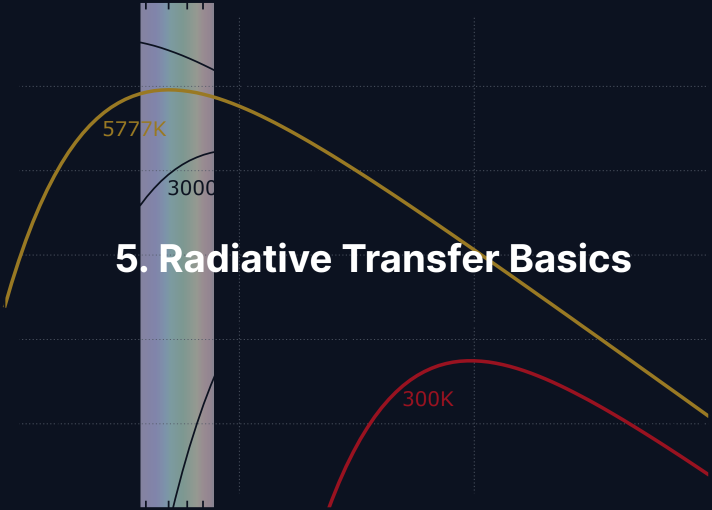
Titan: Thick troposphere (~40 km). Stratosphere heated by organic haze. Extended thermosphere reaches ~1500 km (weak gravity).

Break



Earthrise, Apollo 8 (1968). Credit: NASA

5. Radiative Transfer Basics



Blackbody spectra. Credit: Wikimedia, public domain

Three Interactions: Radiation and Matter

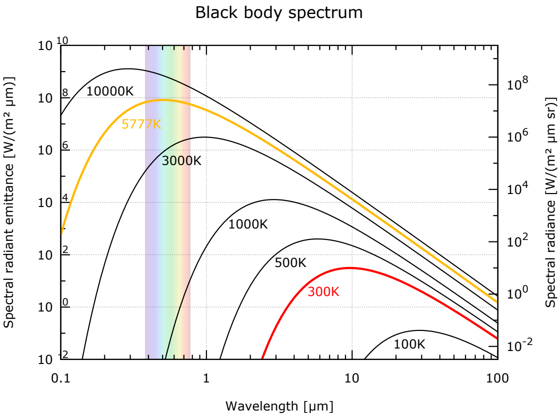
When radiation passes through an atmosphere:

Absorption: Photon energy → internal energy (vibrational, rotational)
Key absorbers: CO_2, H_2O, CH_4, O_3, N_2O

Emission: Warm gas radiates at wavelengths where it absorbs (Kirchhoff's law)
This is how the atmosphere radiates energy to space

Scattering: Photons redirected, not absorbed
Rayleigh ($\propto \lambda^{-4}$): blue sky, red sunsets
Mie (larger particles): less wavelength-dependent

Stellar vs. Planetary Emission



Credit: Wikimedia, public domain

- Sun (~5800 K): peak at ~0.5 μm (visible)
- Planet (~300 K): peak at ~10 μm (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.

Optical Depth

The cumulative absorption along a path through the atmosphere:

$$\tau = \int_0^s \kappa\,\rho\,\mathrm{d}s'$$

- κ = mass absorption coefficient ($m^2\,kg^{-1}$)
- τ is dimensionless: counts "e-folding lengths" of absorption

Optically thin ($\tau \ll 1$):
Most radiation passes through.

Optically thick ($\tau \gg 1$):
Radiation strongly absorbed; only photons from near the "surface" escape.

The Classical Habitable Zone

Where around a star can a planet maintain liquid water?

- Inner edge: runaway greenhouse limit (water fully evaporates)
- Outer edge: CO_2 condenses into ice clouds (Maximum Greenhouse limit)
- For the Sun: classical HZ is ~0.95–1.67 AU (Kopparapu et al. 2013)
- Assumes an Earth-like atmosphere with CO_2/H_2O feedback
- Does not account for tidal heating, magnetic fields, or greenhouse gas diversity
- A starting point, not the last word

Source: Kasting et al. (1993); Kopparapu et al. (2013)

The Moist Adiabatic Lapse Rate

When a parcel rises and condensation occurs, latent heat release warms it relative to the dry adiabat.

$$\Gamma_{\mathrm{moist}} = \Gamma_{\mathrm{dry}} \cdot \frac{1 + \dfrac{L_v\, r_s}{R_d\, T}}{1 + \dfrac{L_v^2\, r_s}{c_p\, R_v\, T^2}}$$

- r_s: saturation mixing ratio; L_v: latent heat of vaporisation
- Earth's lower troposphere: $\Gamma_{\mathrm{moist}} \approx 6$ K/km (compared to 9.8 K/km dry)
- Moist convection is **deeper and more vigorous** than dry; drives thunderstorms and hurricanes
- Latent heat release in rising parcels provides the buoyancy that makes tropical convection reach the tropopause
- Runaway greenhouse reduces Γ further, letting more IR escape to

Deriving the Scale Height (Live)

Template for the blackboard derivation.

1. Start: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho g$ (hydrostatic equilibrium)
2. Ideal gas: $P = \dfrac{\rho\, k_B T}{\mu\, m_u}$
3. Rearrange for ρ: $\rho = \dfrac{\mu m_u P}{k_B T}$
4. Substitute: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\dfrac{\mu m_u g}{k_B T} P$
5. Separate variables and integrate:

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right), \quad H = \frac{k_B T}{\mu m_u g}$$

6. Numerical check: Earth ($T = 288$ K, $\mu = 29$, $g = 9.81$) gives $H \approx 8.4$ km

Source: Catling & Kasting (2017)

Transmission Spectroscopy: Preview

During a transit, starlight passes through the planet's **limb**.

- Absorbing molecules imprint dark features in the transmission spectrum
- Path length ~$2\sqrt{2R_pH}$ through a layer of scale height H
- Signal: $\Delta(R_p/R_\star)^2 \sim 2R_pH/R_\star^2$
- Bigger signal for puffy atmospheres (large H) around small stars
- JWST has detected H_2O, CO_2, CH_4, SO_2 in multiple exoplanet atmospheres
- Full treatment in Lecture 13 (Exoplanet Atmospheres)

Source: Seager & Sasselov (2000); Kreidberg et al. (2014)

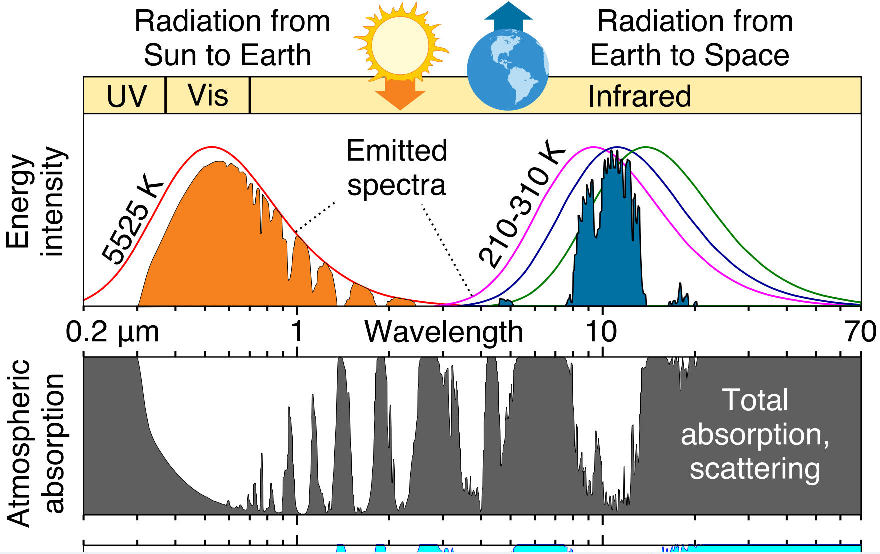
The Beer–Lambert Law

For a beam of initial intensity I_0 through a medium of optical depth τ:

$$\boxed{I = I_0 \, e^{-\tau}}$$

- Intensity drops by factor e per unit optical depth
- **Atmospheric photosphere:** at $\tau \approx 1$, photons escape to space
- Below $\tau = 1$: photons reabsorbed before escaping
- Above $\tau = 1$: photons escape freely
- Temperature at the $\tau \approx 1$ level sets T_{eff}

Earth’s Atmospheric Absorption Spectrum



6. Energy Balance & Greenhouse Effect

Greenhouse effect. Credit: Wikimedia, public domain

Radiative-Convective Equilibrium

A realistic atmosphere combines two regimes:

- **Troposphere:** strong greenhouse absorption → unstable to convection
  - Temperature follows the adiabatic lapse rate
  - Most of the atmospheric mass
  - Clouds and weather occur here
- **Stratosphere:** optically thin → radiative equilibrium
  - Temperature set by the balance of absorbed vs. emitted radiation
  - Stable against convection
  - On Earth, warmed by ozone UV absorption

The boundary (the tropopause) occurs where the radiative gradient becomes less steep than the adiabat.

Manabe and Strickler (1964)

The first realistic 1D radiative-convective model of Earth's atmosphere.

- Solved for the vertical temperature profile with absorbing gases (H_2O, CO_2, O_3)
- Reproduced the surface temperature and the tropopause self-consistently
- Quantified the effect of doubling CO_2: ~2 K warming (first "climate sensitivity" estimate)
- Foundation of modern climate modelling
- Manabe received the 2021 Nobel Prize in Physics for this work

Source: Manabe & Strickler (1964), *J. Atmos. Sci.*

Planetary Energy Balance

Absorbed stellar power = emitted thermal radiation:

$$(1 - A)\frac{L_\star}{4\pi d^2}\,\pi R_p^2 = 4\pi R_p^2\,\sigma\,T_{\rm eff}^4$$

- A = Bond albedo (fraction of light reflected)
- πR_p^2 = cross-sectional area (intercepts light)
- $4\pi R_p^2$ = total surface area (emits isotropically)

Solving:

$$\boxed{T_{\rm eff} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\,d^2}\right]^{1/4}}$$

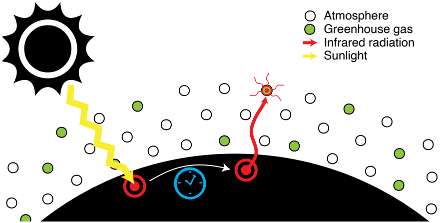
Effective vs. Surface Temperature

Body	A	d **(AU)**	T_{eff} **(K)**	T_{surf} **(K)**	ΔT **(K)**
Venus	0.77	0.72	227	735	+508
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	210	~0
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\mathrm{surf}} - T_{\mathrm{eff}}$ reveals the **greenhouse effect**.
Venus: 508 K. Earth: 33 K. Mars: ~0 K (too thin).

The Greenhouse Mechanism



Credit: Wikimedia, public domain

1. Sunlight (visible, ~0.5 μm) reaches the surface
2. Surface emits thermal IR (~10–15 μm)
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

Result: $T_{\rm surf} > T_{\rm eff}$

One-Layer Greenhouse Model: Setup

Single isothermal layer, emissivity ε in IR, transparent at visible:

Atmospheric balance:

$$\varepsilon\,\sigma\,T_s^4 = 2\,\varepsilon\,\sigma\,T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{T_s^4}{2}$$

Surface balance:

$$(1 - A)\frac{F_\star}{4} + \varepsilon\,\sigma\,T_a^4 = \sigma\,T_s^4$$

Top-of-atmosphere:

$$(1 - A)\frac{F_\star}{4} = \sigma\,T_{\text{eff}}^4$$

One-Layer Greenhouse Model: Result

Combining the three balance equations:

$$\boxed{T_s = T_{\rm eff}\left(\frac{2}{2-\varepsilon}\right)^{1/4}}$$

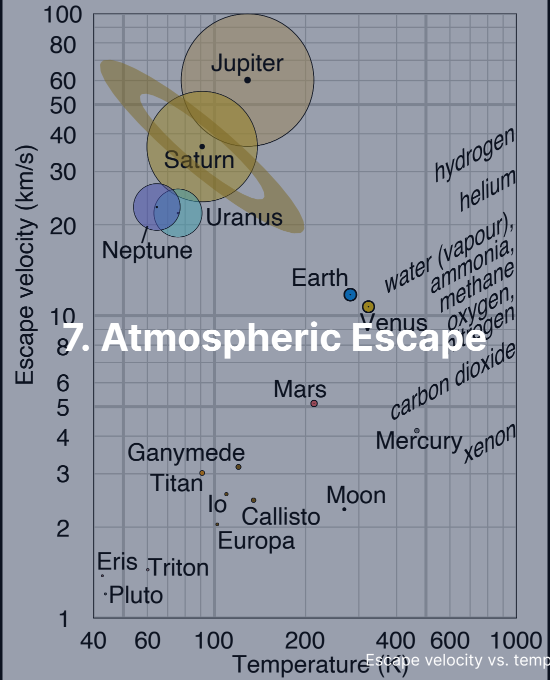
$\varepsilon = 0$ (no greenhouse): $\quad T_s = T_{\rm eff}$

$\varepsilon = 1$ (perfect absorber): $\quad T_s = 2^{1/4}\,T_{\rm eff} \approx 1.19\,T_{\rm eff}$

For Earth: $T_s \approx 1.19 \times 255 \approx 303$ K. Reasonable first estimate.

Next lecture: What happens when T_s rises so high that outgoing radiation saturates? → *Runaway greenhouse* (Lecture 6).

7. Atmospheric Escape



Thermal (Jeans) Escape

Molecules have a Maxwell–Boltzmann velocity distribution. The high-velocity tail extends beyond escape speed.

The **Jeans escape parameter** at the exobase:

$$\lambda_J = \frac{G M_p\, m}{k_{\mathrm{B}} T_{\mathrm{exo}}\, r_{\mathrm{exo}}} = \frac{v_{\mathrm{esc}}^2}{v_{\mathrm{th}}^2}$$

- $\lambda_J \gg 1$: escape negligible (tail too depleted)
- $\lambda_J \lesssim 2$–3: rapid atmospheric erosion
- Escape flux: $\Phi_J \propto (1 + \lambda_J)\, e^{-\lambda_J}$, exponentially sensitive

Jeans Parameter for Earth and Mars

Exobase temperatures: 1000 K (Earth), 300 K (Mars)

Species	m **(u)**	λ_J **(Earth)**	λ_J **(Mars)**
H	1	7.5	5.4
H_2	2	15	11
He	4	30	22
N_2	28	210	150
CO_2	44	330	240

Heavy species (N_2, CO_2): λ_J so large that Jeans escape is negligible.
Atomic H: moderate λ_J → both Earth and Mars lose hydrogen to space.

Hydrodynamic Escape

When stellar EUV heating is intense, escape transitions from molecule-by-molecule to a bulk **atmospheric wind**:

- Most important in the first ~100 Myr (young stars: 10–100× more EUV)
- Can strip $\mathrm{H_2}$-rich primary atmospheres from planets up to several $M_\oplus$
- Outflowing H can **drag along heavier species** (He, C, N, O)
- Leading explanation for the exoplanet **radius valley** (deficit at ~1.5–2 $R_\oplus$; Lecture 13)

Stripping Primary Atmospheres

Young stars are highly active in EUV and X-ray; primordial H/He envelopes can evaporate.

- EUV flux is ~100× higher in the first 100 Myr than today
- Mass loss rate for hydrodynamic regime: $\dot{M} \propto F_{\mathrm{EUV}}\, R_p^3/(GM_p)$
- For sub-Neptunes ($M_p \lesssim 10\, M_\oplus$), atmospheres can be fully stripped in a few hundred Myr
- For Neptune-mass planets and above, envelopes are retained
- Produces the "evaporation valley" at ~1.8 $R_\oplus$ in Kepler data

Source: Owen & Wu (2013); Lopez & Fortney (2013); Fulton et al. (2017)

Energy-Limited Escape Rate

A convenient estimate of mass loss:

$$\dot{M}_{\mathrm{EL}} = \eta \frac{\pi F_{\mathrm{EUV}} R_p^3}{G M_p}$$

- F_{EUV}: incident EUV flux
- η: heating efficiency, typically 0.1–0.2
- Scales as R_p^3: large, fluffy atmospheres lose mass fastest
- Scales inversely with M_p: low gravity eases escape
- Valid when recombination is negligible (“energy-limited regime”)

Source: Watson et al. (1981); Owen (2019)

Non-Thermal Escape Mechanisms

Sputtering: Solar wind ions impact upper atmosphere, eject molecules. Significant for Mars (no global magnetic field).

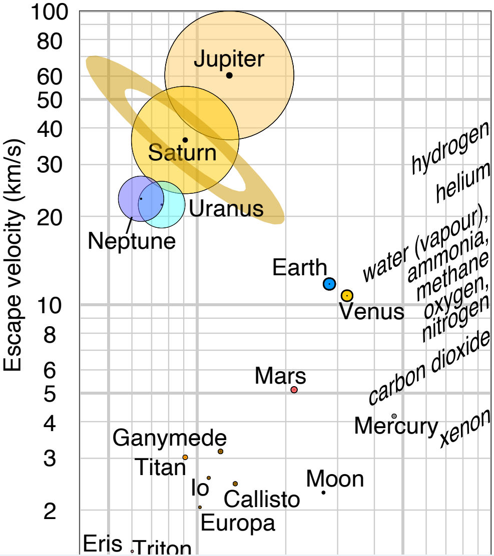
Photochemical: UV dissociates $H_2O \rightarrow$ H + OH; fast H escapes. Key for Venus and Mars water loss.

Ion pickup: Atmospheric atoms ionised by UV, swept away by solar wind B-field. Effective at unmagnetised planets.

Impact erosion: Large impacts can eject substantial atmospheric mass in a single event.

MAVEN at Mars: present loss rate ~100 g s^{-1} (O^+), dominated by solar wind stripping.

Atmospheric Retention: v_{esc} vs. T Diagram

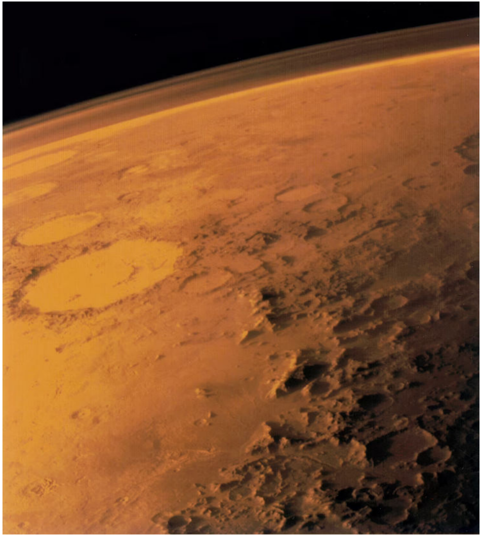


Rule of thumb for long-term retention:

$$v_{\mathrm{esc}} \gtrsim 6\, v_{\mathrm{th}}$$

- Gas giants: retain **everything**
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold → retains N_2 despite low v_{esc}

Mars: A Lost Atmosphere



Credit: NASA/JPL (Viking 1, 1976)

- Present surface pressure: only 6 mbar
- Too thin for liquid water or significant greenhouse warming
- Geological evidence: ancient rivers, lakes, possibly ocean
- Mars had a much thicker atmosphere >4 Gyr ago
- Dynamo loss (~4.1 Ga, Lecture 4) → unshielded solar wind erosion

8. Recent Advances & Outlook

Earth’s atmosphere. Credit: NASA/ISS

Recent Advances

- **JWST / TRAPPIST-1 b:** Thermal emission consistent with bare rock, no significant atmosphere. Inner rocky planets around active M dwarfs may be stripped.
- **JWST / TRAPPIST-1 c:** Similar result, reinforcing theoretical predictions of enhanced atmospheric escape around low-mass stars.
- **MAVEN at Mars:** Quantified loss rates for multiple species. Ion escape (solar wind stripping) dominates over Jeans escape. Integrated losses can account for a substantial fraction of Mars's early atmosphere.

Central theme: atmospheric retention depends on mass, temperature, stellar radiation, magnetic shielding, and geological activity.

Impact Erosion of Atmospheres

Giant impacts can eject substantial atmospheric mass in a single event.

- Shockwaves from a >km-sized impactor accelerate upper-atmospheric gas above escape velocity
- Efficiency depends on impact angle, velocity, and atmosphere mass
- A grazing giant impact can eject the entire atmosphere above the impact horizon
- Key process during the first ~100 Myr of solar system history
- May explain why Earth's early atmosphere was lost despite adequate gravity
- Signature: noble gas isotope ratios ($^{36}\mathrm{Ar}/^{38}\mathrm{Ar}$) record impact-driven loss

Source: Schlichting & Mukhopadhyay (2018); Genda & Abe (2005)

Outgassing vs. Escape: A Dynamic Balance

An atmosphere is not a static inheritance; it is a reservoir in flow.

- **Supply:** volcanic outgassing, impact delivery, biological fluxes
- **Loss:** thermal escape, non-thermal escape, chemical weathering, sequestration in oceans/crust
- The **steady state** depends on the balance of these fluxes
- Earth's active volcanism continually replaces CO_2 on the carbonate-silicate timescale (~0.5 Myr)
- Mars lost its active volcanism and dynamo, so escape was not replenished
- Venus has active (perhaps episodic) volcanism, but no oceans to absorb CO_2

Atmospheric longevity is as much about **replacement** as about initial inventory.

MAVEN at Mars: Quantifying Escape

NASA's MAVEN (2014–) has measured Mars' atmospheric loss in situ.

- Present-day total escape rate: ~1–2 kg/s (all species)
- Dominant loss channel: photochemical escape of oxygen
- Enhanced during solar storms (~10× increase)
- Integrated loss over 4 Gyr: consistent with 0.5–0.8 bar of CO_2 lost
- Combined with Jeans escape of hydrogen: ~15–25 m global equivalent layer of water lost

Source: Jakosky et al. (2018), *Icarus*; Lillis et al. (2017)

JWST and the Fate of Rocky Atmospheres

JWST has begun to directly test whether rocky planets around M dwarfs retain atmospheres.

- TRAPPIST-1 b: MIRI thermal emission consistent with a bare rock (dayside 508 K)
- TRAPPIST-1 c: MIRI shows low heat redistribution $\Rightarrow$ thin or no atmosphere
- LHS 3844 b: similar result using Spitzer data
- Implications: inner rocky planets around active M dwarfs lose their atmospheres
- Open question: whether **outer** TRAPPIST-1 planets retain atmospheres
- JWST will target TRAPPIST-1 e, f during Cycle 3–4

Source: Greene et al. (2023); Zieba et al. (2023), *Nature*

Runaway Greenhouse: Preview for Lecture 6

What happens when a water-rich atmosphere gets too hot?

- More water vapour → stronger greenhouse → hotter surface
- Positive feedback: hotter surface → more evaporation → more greenhouse
- At ~360 W/m^2 absorbed, the **Simpson-Nakajima limit** is reached
- Beyond this, oceans fully evaporate and the runaway becomes unstoppable
- Hydrogen then escapes hydrodynamically, leaving a dry CO_2 atmosphere
- Venus' modern state is the end point of this process (Nakajima 1992; Goldblatt 2012)

Full treatment in **Lecture 6: Atmospheres II**.

Summary

1. Three atmosphere types: **primary** (captured H_2/He), **secondary** (outgassed), **tertiary** (biologically modified)
2. Hydrostatic equilibrium: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$; leads to exponential pressure decay
3. Scale height: $H = k_\mathrm{B} T / (\mu\, m_u\, g)$; encodes thermal energy vs. gravity
4. Vertical structure: troposphere, stratosphere, mesosphere, thermosphere
5. Beer–Lambert law: $I = I_0\, e^{-\tau}$; $\tau \approx 1$ defines the emission level
6. Energy balance: $T_\mathrm{eff} \propto [(1 - A)L_\star / d^2]^{1/4}$; greenhouse raises T_s above T_eff
7. Escape: Jeans (λ_J), hydrodynamic, non-thermal; retention requires $v_\mathrm{esc} \gtrsim 6\, v_\mathrm{th}$

Questions?

Next: **Lecture 6. Atmospheres II: Clouds, Weather, & Climate**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`